Horn Clauses

Horn Clauses

- A Horn clause is a clause (disjunction of literals) with at most/exactly one positive literal.
- General Form: $\neg x_1 \vee x_2, \neg x_2 \vee \cdots \neg x_n \vee y$ is a definite clause,
- whereas $\neg x_1 \vee x_2 \vee x_3$ is not a definite clause, because it has two positive clauses.
- When there is exactly one positive literal in the clause, it is called a *positive* (or *definite*) Horn clause.
- When there are no positive literals, the clause is called a *negative* Horn clause.
- In either case, there can be zero negative literals, and so the empty clause is a negative Horn clause.
- A positive Horn clause $[\neg p_1, \ldots, \neg p_n, q]$ can be read as "$if\ p_1\ and\ \ldots\ and\ p_n, then\ q$." or $p_1 \wedge \ldots \wedge\ p_n \rightarrow\ q$

Types of Horn Clauses

- Horn Clauses come in **three main types**, each serving a different purpose in logic programming and knowledge representation:

1. Facts

No body, only head

Form: B

Example:

```
sunny
Rain
```

provide your base data

2. Rules

Has body and head

Form: $A_1 \wedge \ldots \wedge A_{\square} \rightarrow B$

Example:

```
wet ← rain
happy ← sunny ∧ warm
```

allow inference and reasoning

3. Goals

No head, only body

Form: $\leftarrow A_1 \wedge \ldots \wedge A_{\square}$

Example:

```
← wet
← happy∧sunny
```

drive the computation/query process

Types of Horn Clauses

- **Facts (Unit Clauses):**
  - **Structure:** Just a single positive literal with no conditions
  - **Form:** B
  - **Logical meaning:** "B is true“
  - **No body, only head**
  - **Purpose:** Facts represent unconditional truths in your knowledge base. They're the basic building blocks of what you know to be true.
  - **Examples:**
    - bird(sparrow)
    - Sunny
    - parent(john, mary)
    - student(alice)
    - Happy(john) (We may not claim who john is or what happy does mean, but we may agree that The sentence “Happy(john)” claims that the individual named by “john” (whoever that might be) has the property named by “Happy” (whatever that might be))

Types of Horn Clauses

- Rules (Definite Clauses):
  - **Structure:** One positive literal (head) with one or more conditions (body)
  - **Form:** $B \leftarrow A_1 \land A_2 \land \ldots \land A_n$
  - **Or:** $A_1 \land A_2 \land \ldots \land A_n \rightarrow B$
  - **Meaning:** "B is true IF all of $A_1, A_2, \ldots, A_n$ are true"
  - **Purpose:** Rules define relationships and derive new knowledge from existing facts. They're the reasoning component of your knowledge base.
  - **Examples:**
  - flies(X) ← bird(X) ("X flies if X is a bird")
  - grandparent(X, Z) ← parent(X, Y) ∧ parent(Y, Z). ("X is grandparent of Z if X is parent of Y AND Y is parent of Z")
  - happy(X) ← sunny ∧ warm. ("X is happy if it's sunny AND warm")
  - pass(Student) ← attendance(Student, Good) ∧ grade(Student, Pass) ("Student passes if they have good attendance AND passing grade")

Types of Horn Clauses

- Goals (Queries):
  - **Structure:** Only negative literals, no positive head
  - **Form:** $\leftarrow A_1 \land A_2 \land \ldots \land A_{\square}$
  - **Meaning:** “Is it true that A_1 AND A_2 AND ... AND $A_{\square}$?”
  - **Purpose:** Goals are questions we ask the knowledge base. The system tries to prove the goal is true using available facts and rules.
  - **Examples:**
    - ← flies(sparrow). ("Does sparrow fly?")
    - ← grandparent(john, alice). ("Is john a grandparent of alice?”)
    - ← parent(X, mary). ("Who is a parent of mary?" (finds all X))
    - ← happy(Person) ∧ student(Person). ("Find all persons who are both happy and students“)

Types of Horn Clauses

Facts	Rules	Goals
bird(sparrow). bird(eagle). bird(penguin). mammal(cat). mammal(dog).	flies(X) ← bird(X) ∧ ¬penguin(X). animal(X) ← bird(X). animal(X) ← mammal(X). pet(X) ← mammal(X) ∧ domesticated(X).	← flies(sparrow) ← flies(cat). ← animal(eagle). ← animal(X).

Knowledge Base:

Facts:

```
bird(sparrow)
bird(eagle)
mammal(cat)
```

Rules:

```
flies(X) ← bird(X)
animal(X) ← bird(X)
animal(X) ← mammal(X)
```

Queries (Goals):

```
← flies(sparrow)
```
✓ True

```
← flies(cat)
```
✗ False

```
← animal(eagle)
```
✓ True

```
← animal(cat)
```
✓ True

Why Horn Clauses Matter

Computational Efficiency

Linear time resolution using forward/backward chaining

Decidability

Satisfiability of Horn clauses is decidable in polynomial time

Foundation for Logic Programming

Prolog and similar languages use Horn clause logic

Natural Knowledge Representation

Easy to express rules and facts in a declarative way

Expressive Power

Can represent many real-world scenarios and relationships

Applications in AI & Knowledge Representation

Expert Systems

Medical diagnosis, fault detection, and decision support systems

Semantic Web & Ontologies

Knowledge graphs, reasoning about relationships and properties

Database Query Optimization

Datalog and deductive databases use Horn clause logic

Natural Language Processing

Grammar rules and semantic analysis using definite clauses

Identify Horn Clause Types

- student(alice).
- $\leftarrow$ enrolled(X, Course).
- passes(X) $\leftarrow$ studies(X) $\wedge$ attends(X).
- temperature(high).
- $\leftarrow$ parent(john, mary).
- sibling(X, Y) $\leftarrow$ parent(Z, X) $\wedge$ parent(Z, Y) $\wedge$ X $\neq$ Y.
- $\leftarrow$ happy(Person) $\wedge$ wealthy(Person).
- raining.